Thm: If $u \in W^{k,p}(U)$ and $\zeta \in C_c^\infty(U)$, then

$$D(\zeta u) = \zeta Du + u D\zeta.$$

Pf: For $\varphi \in C_c^\infty(U)$,

$$\int_U \zeta u D\varphi = \int_U u\left(D(\overbrace{\zeta\varphi}^{\in C_c^\infty(U)}) - \varphi D\zeta\right)$$

$D(\zeta\varphi) = \zeta D\varphi + \varphi D\zeta$

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u$$

$$= \int_U u\, D(\zeta\varphi) - \int_U u\varphi\, D\zeta$$

$$= -\int_U \zeta\varphi\, Du - \int_U u\varphi\, D\zeta$$

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u$$

$$\int \zeta u D^\alpha \varphi = -\int_U (\zeta Du + u D\zeta)\varphi \quad \forall \varphi \in C_c^\infty \#$$

Recall: $u_j \to u$ in $W^{k,p}(U)$ if $\|u_j - u\|_{W^{k,p}(U)} \to 0$.

Recall: $V \subset\subset U$ means $\overline{V}$ is compact and is contained in U.

Def: $u_j \to u$ in $W^{k,p}_{loc}(U)$ if $u_j \to u$ in $W^{k,p}(V)$ for every $V \subset\subset U$.

Notation: We write $W^{k,2}(U) = H^k(U)$ (so $H^0(U) = L^2(U)$).

<u>Def:</u> We define the subspace $W_0^{k,p}(U)$ of $W^{k,p}(U)$ to be the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$, i.e., $u \in W^{k,p}(U)$ is in $W_0^{k,p}(U)$ iff $\exists$ a seq (u_j) in $C_c^\infty(U)$ st $u_j \to u$ in $W^{k,p}(U)$. We write $W_0^{k,2}(U) = H_0^k(U)$.

Approximation by Smooth Functions

Recall:

$$\eta(x) = \begin{cases} C e^{-\frac{1}{1-|x|^2}} & \text{if } |x|<1 \\ 0 & \text{if } |x| \geq 1 \end{cases}$$

$C>0$ is chosen so that $\int_{\mathbb{R}^n} \eta(x)\,dx = 1$.

$\eta \in C^\infty(\mathbb{R}^n)$, $\operatorname{spt} \eta = \overline{B_1(0)}$

$$\int_{\mathbb{R}^n} \eta(x)\,dx = C \int_{B_1(0)} e^{-\frac{1}{1-|x|^2}}\,dx$$

For $\varepsilon > 0$, let $\eta_\varepsilon(x) = \frac{1}{\varepsilon^n}\eta\left(\frac{|x|}{\varepsilon}\right)$. Then $\operatorname{spt}\eta_\varepsilon = \overline{B_\varepsilon(0)}$

and $\int_{\mathbb{R}^n} \eta_\varepsilon(x)\,dx = 1$. Let $U \subset \mathbb{R}^n$ be an open

set. For a function $u \in L^1_{loc}(U)$, define

$$u_\varepsilon = \eta_\varepsilon * u,$$

i.e.,

$$u_\varepsilon(x) = \int_U \eta_\varepsilon(x-y)\,u(y)\,dy, \quad x \in U_\varepsilon := \{u \in U : \operatorname{dist}(x, \partial U) > \varepsilon\}$$

(mollifier).

$$\int_{\mathbb{R}^n} \eta_\varepsilon(x)\,dx = \frac{1}{\varepsilon^n}\int_{\mathbb{R}^n} \eta\left(\frac{|x|}{\varepsilon}\right)dx$$

$x = \varepsilon y$, $dx = \varepsilon^n dy$

$$= \int_{\mathbb{R}^n} \eta(y)\,dy = 1$$

and $u_\varepsilon \to u$ a.e. in U

Then $u \in C^\infty(U_\varepsilon)$. If $u \in L^p_{loc}(U)$,

then $u_\varepsilon \to u$ as $\varepsilon \to 0$ in

$L^p_{loc}(U)$, i.e., $u_\varepsilon \to u$ as $\varepsilon \to 0$

in $L^p(V)$ for any $V \subset\subset U$.

$\operatorname{spt} \eta_\varepsilon(x - \cdot) = \overline{B_\varepsilon(x)}$

$U_\varepsilon = \{x \in U : \operatorname{dist}(x, \partial U) > \varepsilon\}$

Thm: If $1 \le p < \infty$, $u \in W^{k,p}(U)$, then $u_\varepsilon \to u$ as $\varepsilon \to 0$ in $W^{k,p}_{loc}(U)$.

$\in C^\infty(U)$

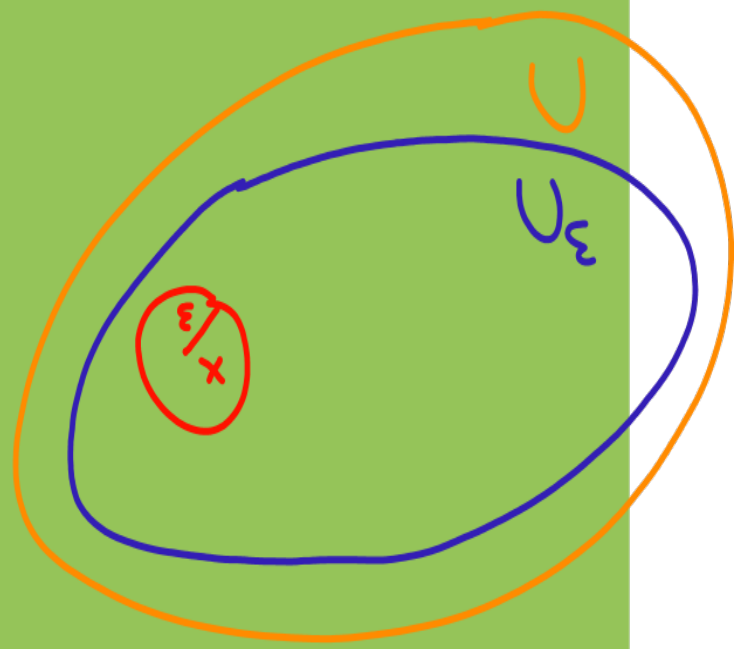
Pf: First we show that $\underbrace{D^\alpha u_\varepsilon}_{\text{usual } \alpha\text{th partial derivative of } u_\varepsilon} = (D^\alpha u)_\varepsilon = \eta_\varepsilon * D^\alpha u$ for each multi-index α with $|\alpha| \le k$.

For fixed $x \in U$ and $\varepsilon > 0$ so small that $x \in U_\varepsilon$,

$$u_\varepsilon(x) = \int_U \eta_\varepsilon(x-y)\, u(y)\, dy,$$

so

$$D^\alpha u_\varepsilon(x) = \int_U u(y)\, \underbrace{D_x^\alpha \eta_\varepsilon(x-y)}_{(-1)^{|\alpha|} D_y^\alpha \eta_\varepsilon(x-y)}\, dy$$



$$= (-1)^{|\alpha|} \int_U u(y)\, D_y^\alpha \underbrace{\eta_\varepsilon(x-y)}_{\in C_c^\infty(U)}\, dy$$

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u$$

$$= \int_U \eta_\varepsilon(x-y)\, D^\alpha u(y)\, dy$$

$$= (\eta_\varepsilon * D^\alpha u)(x)$$

$$= (D^\alpha u)_\varepsilon(x).$$

Let $V \subset\subset U$. Then for each α with $|\alpha| \le k$, $D^\alpha u_\varepsilon = (D^\alpha u)_\varepsilon$

$\to D^\alpha u$ in $L^p(V)$ as $\varepsilon \to 0$. So

$$\|u_\varepsilon - u\|_{W^{k,p}(V)} = \left(\sum_{|\alpha| \le k} \|D^\alpha (u_\varepsilon - u)\|^p_{L^p(V)} \right)^{1/p}$$

$$= \left(\sum_{|\alpha| \le k} \|D^\alpha u_\varepsilon - D^\alpha u\|^p_{L^p(V)} \right)^{1/p} \to 0 \text{ as } \varepsilon \to 0. \#$$

Thm: If $1 \le p < \infty$ and U is bdd, then for any $u \in W^{k,p}(U)$, $\exists$ a sequence (u_j) in $W^{k,p}(U) \cap C^\infty(U)$ st $u_j \to u$ in $W^{k,p}(U)$.

Thm: If $1 \le p < \infty$, U is bdd, and $\partial U \in C^1$, then $\exists$ a seq (u_j) in $C^\infty(\bar{U})$ st $u_j \to u$ in $W^{k,p}(U)$.